Metaplectic Borel No-Flip

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Summary

Pass 80 tests whether the finite Weil/metaplectic symmetry descends to the solid phantom. At every finite level the usual Weyl flip exists, but the solid limit has no map $\epsilon \to \mathbb{Q}$. The surviving symmetry is only the Borel group

$$B = \mathbb{Q}^\times \ltimes \epsilon,$$

the affine group preserving the ϵ polarization.

Main Claims

- **Finite levels are fully symplectic.** For $\mathbb{Z}/N$, the group $\mathrm{SL}_2(\mathbb{Z}/N)$ has the expected Bruhat decomposition, Weyl element, and finite Fourier transform.
- **Limit wall.** The lower-left entry needed for the Weyl flip would be a solid morphism $\epsilon \to \mathbb{Q}$, but $\mathrm{Hom}_{\mathrm{Solid}}(\epsilon, \mathbb{Q}) = 0$.
- **No metaplectic descent.** The finite Fourier transforms F_N are unitary, satisfy $F_N^4 = I$, and have Gauss sums of the expected norm, but their candidate limit lies in the vanished Hom group.

Machine Verification

`code/scripts/check-pass80.py` produced `artifacts/reports/pass80-metaplectic-borel-noflip-check.json` with overall PASS. It verifies finite SL_2 order/Bruhat identities, Weyl-flip properties, the one-sided Hom tower, and the finite Weil transform checks.

Next Use

The pass identifies the representation-theoretic wall for the next stage: the phantom has an affine/Borel action, but no Weyl operator and hence no self-dual Fourier functional equation.